Note 5.1 Area and Estimating with Finite Sums

Area

- **Upper Sum, Lower Sum and Midpoint Rule:**
    - **Strategy to get the area of an irregular graph:** To split the graph into many same-width rectangle and approximate the area with finite sums.
    - **Upper Sum:** Taking the height of each rectangle as the maximum value of $f(x)$ for a point x in the base interval of the rectangle.
    - **Lower Sum:** Taking the height of each rectangle as the minimum value of $f(x)$ for a point x in the base interval of the rectangle.
    - **Midpoint Rule:** Taking the height of each rectangle as the value of the midpoint of $f(x)$ for the base interval of the rectangle.
- **Finite sums:** If we take the interval $[a, b]$ into n subintervals with equal width $\Delta x = \frac{b-a}{n}$ and f is evaluated at the points in each subintervals: c_1 in the first subinterval, c_2 in the second subinterval and so on. Then the finite sums is

$$f(c_1)\Delta x + f(c_2)\Delta x + \cdots + f(c_n)\Delta x.$$

Distance Traveled

- **Strategy:** Suppose that the velocity function $v(t)$ has been given and the direction doesn't change. If we already know the antiderivative $F(t)$ of $v(t)$, we can find the position function $s(t) = F(t) + C$ thus the distance in the interval $[a, b]$ will be $s(b) - s(a) = F(b) - F(a)$. If we don't know the antiderivatives, we can subdivide the interval $[a, b]$ into n subintervals with the same width Δt, then the sum of distances traveled over $[a, b]$ is

$$D \approx v(t_1)\Delta t + v(t_2)\Delta t + \cdots + v(t_n)\Delta t$$

Displacement Versus Distance Traveled

- **Displacement:** The difference between its initial and final positions($s(b) - s(a)$). And as above, subdivide the interval $[a, b]$ into n subintervals with the same width Δt, then the displacement of the interval $[t_{k-1}, t_k]$ is

$$v(t_k)\Delta t.$$

  in which $v(t_k)$ is the approximation of the velocity in the interval and it can be positive or negative.

- **Distance Traveled:** As above, the distance traveled of the $[t_{k-1}, t_k]$ is

$$|v(t_k)|\Delta t$$

and the total distance traveled is approximately the sum

$$|v(t_1)|\Delta t + |v(t_2)|\Delta t + \cdots + |v(t_n)|\Delta t.$$

Average Value of a Nonnegative Continuous Function

- **Definition:** The area under the function graph on the interval $[a, b]$ divides by $b - a$ to get the average value of a nonnegative continuous function on the interval $[a, b]$.